Hanu Skanda Banappa

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Professional Summary

Electrical Engineering PhD candidate specializing in neural signal processing and closed-loop control systems. Proven track record in developing high-fidelity machine learning and brain connectivity models for intracranial EEG (iEEG) recordings. Experienced in collaborating directly with neurosurgeons and clinicians in clinical settings to brainstorm and refine novel measures for epilepsy research. Seeking an R&D internship in neurotechnology to translate complex neural data into deployable neural interface systems.

Skills

Neural Signal Processing
SEEG/EEG/EMG pipelines | Spectral analysis | Brain connectivity analysis | Feature engineering | Machine learning architectures

Control Systems
Classical & Modern control theory | Active Disturbance Rejection Control | System modeling and stability analysis

Scientific Computing
Python | MATLAB | Simulink | High-performance computing | Git

Instrumentation & Systems
Real-time biosensing | Data acquisition | Sensor interfacing

Projects

Seizure Onset zone Localization Using AI (iEEG)

- Developed a transformer-based deep learning model to identify seizure onset zones from intracranial EEG recordings.
- Addressed severe class imbalance in seizure onset detection to improve robustness and clinical relevance
- Implemented explainable visualization tools to interpret learned model features and support clinical validation.

Tracking Brain Connectivity (iEEG)

- Built functional connectivity analysis pipelines for multi-channel intracranial EEG data.
- Applied causal and non-causal connectivity metrics to characterize seizure network dynamics.
- Analyzed high-temporal-resolution microstate transitions to identify connectivity-based biomarkers for epileptic seizures.

Real-Time EMG Biosensing device (EMG)

- Designed and built a real-time electromyography acquisition system using custom differential amplifiers, analog filters, and Arduino-MATLAB based data acquisition.
- Implemented signal preprocessing pipelines for noise reduction and muscle activation detection in real time.

Automatic steering of steel mill (ADRC)

- Developed and deployed an Active Disturbance Rejection Control (ADRC) system for automated steering in an industrial steel mill application provided by Integrated Milling Systems Inc.
- Modeled plant dynamics and validated controller performance through simulation.
- Demonstrated improved robustness and reduced tuning efforts compared to conventional PID-based controllers
- Worked with PLC environments (Rockwell automation platforms) to integrate modern control methods into industrial workflows.

Work Experience

Graduate Researcher – Neural Engineering January 2023 - Present
Cleveland Clinic, Department of Epilepsy, Cleveland, USA

- Built and curated large-scale intracranial EEG datasets in standardized formats, enabling reuse across multiple research pipelines within the department.
- Developed signal processing and machine learning pipelines for seizure detection, localization, and characterization of neural network dynamics.
- Designed and validated connectivity-based network biomarkers to characterize seizure and non-seizure neural dynamics.
- Implemented explainable AI to support interpretation of learned features from deep learning models trained on neural data.
- Collaborated with clinicians and engineers to translate clinical questions into quantitative engineering models.

Hardware Test Engineer & Program Manager January 2022 – July 2022
ElectricPe, Bengaluru, India

- Designed testing protocols for EV charging units, improving efficiency by ~50% prior to field testing deployments
- Led debugging and failure analysis of hardware prototypes and coordinated corrective actions across teams.
- Built tracking and monitoring systems to manage deployed devices and inventory using barcode-based workflows.
- Improved operational efficiency through cross-functional coordination between engineering, testing, and field teams.
- Led and managed a team of four engineers in debugging, troubleshooting, and failure analysis, improving team efficiency by ~40%

Education

Cleveland State University, Cleveland, OH Exp Grad: December 2028
PhD in Electrical Engineering (in Progress) GPA: 3.7/4.0